

- 4 (a) The heating effect is due to power dissipation in the coil which is dependent on the root-mean-square current and not on the average current. [C1]
The root mean square current is the value of the steady direct current that would dissipate heat at the same rate as the alternating current in a given resistor and it is non-zero. Hence, there is heating effect in the coil. [C1]

(b) (i)
$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$
$$V_s = 15 \times 16k = 240kV \quad [M1]$$
$$P_{loss} = I^2 R$$
$$= \left(\frac{20 \times 10^6}{240 \times 10^3} \right)^2 \times 20 \times 10 = 1.39 \times 10^6 W \quad [A1]$$

(ii)
$$P_{loss} = \left(\frac{20 \times 10^6}{16 \times 10^3} \right)^2 \times 20 \times 10 = 3.13 \times 10^8 W$$
$$\text{Energy saved} = (3.13 \times 10^8 - 1.39 \times 10^6) \times 24 \div 1000$$
$$= 7.48 \times 10^6 \text{ kwh}$$
$$\text{Amount saved} = 7.48 \times 10^6 \times 0.10$$
$$= \$7.48 \times 10^5$$